

UNDERSTANDING ROBUSTNESS IN VISION TRANSFORMERS

ABSTRACT. In their 2023 paper *Understanding and Improving Robustness of Vision Transformers through Patch-based Negative Augmentation* [4], Qin et al. conduct an informative study regarding the robustness of Vision Transformer (ViT) models, and propose a novel way of training such models to improve robustness in light of their findings. In this write-up, we summarize the key points of their paper and propose a novel idea of our own which aims to improve the robustness of ViT models on an inherent level, without the need for augmented training.

1. PAPER SUMMARY

1.1. Empirical Findings. To better understand how ViTs learn classifying features of the input data, Qin et al. evaluated Google’s ViT-B/16 on ImageNet-1k data that had been subject to various “patch-based” transformations (see Fig. 1). They found that these transformations were largely immaterial to both the model’s predictions and confidence scores, suggesting that the features of the input data that are learned by ViTs are among those which are preserved by such transformations. The authors furthermore assessed whether this aspect of ViTs had an impact on model robustness by training a copy of ViT-B/16 exclusively on patch-transformed ImageNet-1k data (so as to force the model to fully rely on features of the data preserved by these transformations) and compared its performance to that of a baseline ViT-B/16 model on the (clean) ImageNet-1k validation set and ImageNet-A, ImageNet-C, and ImageNet-R robustness benchmarks. The same training details and hyperparameters were used for both models so as to ensure that the only difference between them was the training images themselves. They found that while the two models performed comparably on the clean validation data, the model trained on patch-transformed data performed worse on the robustness benchmarks. These findings suggest that the features of the input data that are learned by ViTs, while useful (in the sense that they promote high classification accuracy), impede the robustness of the model.

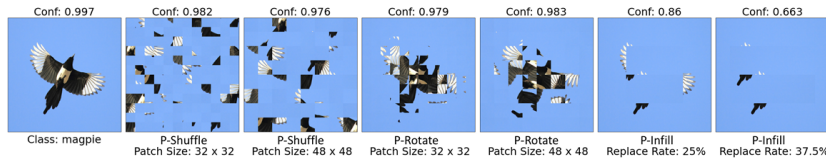


FIGURE 1. Example illustrating ViT-B/16’s lack of sensitivity to the patch-based transformations P-Shuffle (randomly shuffling the positions of the patches), P-Rotate (randomly rotating the patches), and P-Infill (replacing the patches in the interior of the image with those on the boundary)

1.2. Patch-Based Negative Augmentation. To prevent ViTs from relying on these non-robust features, the authors propose a new way of training them, which they dub *patch-based negative augmentation*. The idea is essentially to show the model both clean data as well as patch-transformed data during training, and encourage it towards a particular truth label for data of the latter type. Specifically, given a clean image x , one can apply a patch-based transformation to x to produce its so-called *negative view* $\tilde{x}$. Letting $\mathcal{L}_{ce}(\mathcal{B}; \theta)$ denote the cross-entropy loss function for a ViT with parameters θ evaluated on a minibatch of clean examples $\mathcal{B}$, the authors propose training according to the loss function

$$(1) \quad \mathcal{L}(\mathcal{B}, \tilde{\mathcal{B}}; \theta) := \mathcal{L}_{ce}(\mathcal{B}; \theta) + \lambda \mathcal{L}_{neg}(\tilde{\mathcal{B}}; \theta),$$

where $\tilde{\mathcal{B}}$ is the minibatch of corresponding negative views, $\mathcal{L}_{neg}(\tilde{\mathcal{B}}; \theta)$ is some loss function evaluated specifically on this batch of negative views, and λ is a fixed parameter balancing the importance of clean versus transformed data. There are several ways to choose the loss function $\mathcal{L}_{neg}(\tilde{\mathcal{B}}; \theta)$ on negative views. Two such choices discussed by the authors are the *uniform loss* (in which $\mathcal{L}_{neg}$ is taken to be the cross-entropy loss function but with the one-hot truth labels replaced with uniform distribution labels), and the ℓ^2 *loss* (in which $\mathcal{L}_{neg}$ is taken to be the batch-normalized sum of ℓ^2 distances between the clean logits and corresponding negative view logits). The first choice encourages the model to assign to negative views truth labels which essentially reflect random guesses. The second choice provides an alternative to this in which we do not presuppose a truth label, and instead encourage the logits of the clean examples to be “far away” from the logits of their corresponding negative views. This choice furthermore motivates the choice of terminology *negative augmentation*, as many recent works [2, 5, 1] utilize training of the form (1) to encourage close logits between clean and so-called “positively” augmented counterparts.

1.3. Results. By training two copies of ViT-B/16, one normally and the author via patch-based negative augmentation, the authors observe that the latter leads to a holistic improvement to the robustness of the model. Specifically, the model trained in this way consistently outperformed the other one on the ImageNet-A, ImageNet-C, and ImageNet-R benchmarks, typically on the order of 0.5-1.5%, while performing comparably on clean data.

2. EXPERIMENT

2.1. Motivation. Augmented training, while evidently effective, suffers from the fact that it does not make the model *inherently* more robust; rather, any improvements to robustness stem solely from how the model was trained. To this end, we investigated whether one could achieve results similar to those of Qin et al. by instead making an educated change to the fundamental architecture of ViT models. Specifically, the findings of the paper suggest strongly that ViTs struggle to learn global classifying features of the input data, instead favoring classifying features that are, in some sense, supported in patches. Furthermore, they demonstrate that features of the latter type are non-robust. For this reason, we posit that a change to the fundamental architecture of ViT models which promotes the learning of global classifying features should lead to an improvement in model robustness. The specific change we investigate is the replacement of the patch embedding layer with *two* such layers, one utilizing a small patch size (for accuracy and overall performance) and the other a larger patch size (to encourage the identification of features that are more global).

2.2. Outline & Results. To evaluate the efficacy of this change empirically, we developed two simple ViT models; one with a standard architecture, the “baseline” model, and one with the proposed change, the “dynamic” model. Explicitly, the baseline model consisted of a single patch embedding layer, transformer block, and MLP head, alongside some dropout and average pooling layers to prevent overfitting and accelerate training (see Fig. 2 below). In contrast, the “dynamic” model utilized exactly the same architecture, but with two patch embedding layers instead of one. The outputs of these layers are then run through the model in parallel until passing through the transformer block, at which point they are average pooled along the number of patches and added together so as to combine the information captured by the two embeddings (again, see Fig. 2 below).

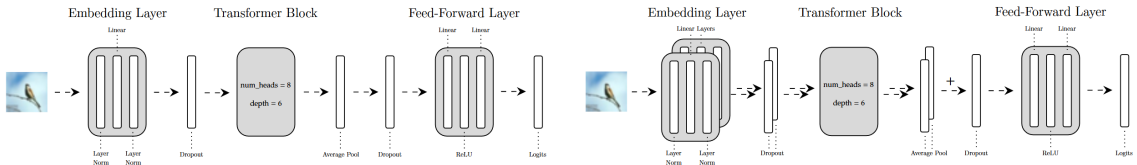


FIGURE 2. Architecture of baseline model (left) versus architecture of dynamic model (right)

The models were then trained CIFAR-10 data for forty epochs with the same training details and hyperparameters, and finally evaluated on clean validation data as well as adversarial data generated using three popular attack algorithms from the `torchattacks` library (namely FGSM, BIM, and PGD; see [3] for details regarding what these entail). We found that the dynamic model performed comparably to the baseline model on clean data while consistently outperforming the baseline model on the robustness benchmarks, especially as the severity of the attack increased. These findings are summarized below in Table 1.

	Train	Test	Attack	$\varepsilon \approx 0.03$	$\varepsilon \approx 0.06$	$\varepsilon \approx 0.09$	$\varepsilon \approx 0.12$
B-ViT	81.28	73.82	FGSM	28.70	18.53	12.52	9.05
			BIM	37.02	19.79	15.11	13.75
			PGD	7.70	2.56	1.73	1.41
D-ViT	78.21	72.15	FGSM	29.53 (+0.83)	19.42 (+0.89)	13.47 (+0.95)	10.05 (+1.00)
			BIM	36.71 (−0.31)	20.40 (+0.61)	15.59 (+0.48)	14.50 (+0.75)
			PGD	7.86 (+0.16)	2.78 (+0.22)	1.83 (+0.10)	1.43 (+0.02)

TABLE 1. Performance of the baseline model (B-ViT) versus the dynamic model (D-ViT) on clean train and test data, and on adversarial data generated using the `torchattacks` library. Here, ε is a parameter which can be roughly thought of as the strength of the attack, with default value $8/255 \approx 0.03$

2.3. Conclusion. Implementing multiple patch embedding layers with differing patch sizes seems to make ViT models inherently more robust, with improvements occurring on a similar scale to those attained by Qin et al. (one should keep in mind, however, that we conducted our experiment on a much simpler dataset). These findings open many avenues for further experimentation. In particular, knowing that dynamic patch embedding layers improve model robustness, what is the optimal way to implement this feature? For instance, in our experiment we utilized two such layers, but perhaps one would see better results with several such layers. Moreover, what is the best way to combine the the information captured by the different patch embeddings? In our case, we added the corresponding outputs of the transformer block, however it is conceivable that some other mechanism of combining them (e.g. concatenation or weighted averaging) could lead to better results.

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